

Joint convexity of the Umegaki quantum relative entropy

A conventional-mathematics proof, in exact correspondence with the Lean formalization `qRelativeEnt_joint_convexity` (PhysLean)

Prepared for review — PhysLean contribution

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We give a self-contained, conventional-mathematics proof that the Umegaki quantum relative entropy is jointly convex. The argument is presented exactly as it is formalized in Lean 4 in the PhysLean library (theorem `qRelativeEnt_joint_convexity`, commit `9385c7f3`): every step below corresponds to an identifiable step in the formal proof, and §7 gives the full paper-to-Lean dictionary. The proof route is the Frank–Lieb / Leditzky–Rouzé–Datta variational argument for the sandwiched Rényi trace functional, taken to the limit $\alpha \rightarrow 1^+$; it deliberately avoids differentiability at $\alpha = 1$, using instead a one-sided convergent majorant (Lemma E, the one genuinely new ingredient) that requires only continuity plus two elementary scalar bounds. The purpose of this paper is to let a reviewer confirm the *mathematical* correctness of the contribution in ordinary language, independently of reading Lean, while §7 makes the correspondence to the machine-checked proof precise.

This note renders the Lean proof of `qRelativeEnt_joint_convexity` (`QuantumInfo/Entropy/DPI.lean`, line ~1483 as of commit `9385c7f3`) as a mathematician would write it on paper. The structure of the argument below follows the Lean proof step for step; Lean identifiers are given in `code font` at the point where they are used. Nothing is included that is not in the formal proof, and the two pre-existing ingredients it relies on (joint convexity of the trace functional, and continuity in α) are stated as propositions with their own provenance.

1. Setting and conventions

Throughout, d is a finite index set and states live on $\mathbb{C}^d$.

- A `state` (`MState d`, `QuantumInfo/States/Mixed/MState.lean:63`) is a Hermitian matrix $\rho \geq 0$ with $\text{Tr } \rho = 1$.
- For $A \geq 0$ and $r \in \mathbb{R}$, the power A^r is defined by functional calculus with $0^r := 0$ for $r \neq 0$ (`HermitianMat.rpow`, `QuantumInfo/ForMathlib/HermitianMat/Rpow.lean:45`); negative powers are therefore Moore–Penrose-style powers on the support. Likewise $\log A$ is taken on the support.
- We write $\text{supp } \rho \subseteq \text{supp } \sigma$ for the support (= range) inclusion of PSD matrices; in Lean this is the kernel inclusion $\sigma.\text{M.ker} \leq \rho.\text{M.ker}$.

- Entropies take values in $[0, \infty]$ ($\mathbb{R}_{\geq 0\infty}$), with the conventions $a \cdot \infty = \infty$ for $a \neq 0$ and $0 \cdot \infty = 0$.
- p ranges over $\text{Prob} = \{p \in \mathbb{R} \mid 0 \leq p \leq 1\}$ (`QuantumInfo/ClassicalInfo/Prob.lean:36`). The mixture notation $p \llbracket \rho_1 \leftrightarrow \rho_2 \rrbracket$ (`Mixable.mix`, `Prob.lean:267`) denotes the state whose matrix is $p\rho_1 + (1-p)\rho_2$. In the theorem statement, $1 - p$ on the right-hand side is truncated subtraction in $\mathbb{R}_{\geq 0\infty}$, which equals the real $1 - p$ since $p \leq 1$.
- $\log$ is the natural logarithm.

2. Definitions

Definition 1 (Umegaki relative entropy; `qRelativeEnt`, notation `D`, `Relative.lean:1497`).

$$D(\rho \parallel \sigma) = \begin{cases} \text{Tr}[\rho(\log \rho - \log \sigma)] & \text{if } \text{supp } \rho \subseteq \text{supp } \sigma, \\ +\infty & \text{otherwise.} \end{cases}$$

(In Lean, $\mathbf{D}(\rho \parallel \sigma)$ is *defined* as $\mathbf{D_1}(\rho \parallel \sigma)$, the $\alpha = 1$ member of the next definition; the identification with the trace formula under the support condition is `qRelativeEnt_ker`, `Relative.lean:1782`.)

Definition 2 (sandwiched Rényi relative entropy; `SandwichedRelRentropy`, notation `Dα`, `Relative.lean:1480`). For $\alpha > 0$, set $\gamma := \frac{1-\alpha}{2\alpha}$. If $\text{supp } \rho \not\subseteq \text{supp } \sigma$, then $\tilde{D}_\alpha(\rho \parallel \sigma) := +\infty$. Otherwise

$$\tilde{D}_\alpha(\rho \parallel \sigma) := \begin{cases} \frac{1}{\alpha-1} \log \text{Tr}[(\sigma^\gamma \rho \sigma^\gamma)^\alpha] & \alpha \neq 1, \\ \text{Tr}[\rho(\log \rho - \log \sigma)] & \alpha = 1, \end{cases}$$

a nonnegative quantity (`sandwichedRelRentropy_nonneg`). (For $\alpha \leq 0$ the Lean definition returns the junk value 0; only $\alpha \geq 1$ is used here.)

Definition 3 (sandwiched trace functional; `sandwichedTraceFunctional`, notation `Qα`, `DPI.lean:66`). For $\alpha \in \mathbb{R}$ and $\gamma = \frac{1-\alpha}{2\alpha}$,

$$\tilde{Q}_\alpha(\rho \parallel \sigma) := \text{Tr}[(\sigma^\gamma \rho \sigma^\gamma)^\alpha] \in \mathbb{R}.$$

For $0 < \alpha \neq 1$ and $\text{supp } \rho \subseteq \text{supp } \sigma$ one has (`sandwichedRelRentropy_eq_log_traceFunctional`, `DPI.lean:78`)

$$\tilde{D}_\alpha(\rho \parallel \sigma) = \left[\frac{1}{\alpha-1} \log \tilde{Q}_\alpha(\rho \parallel \sigma) \right]^+ \quad (\text{as an element of } [0, \infty], \text{ via } \text{ofReal}).$$

3. Theorem

Theorem (`qRelativeEnt_joint_convexity`, `DPI.lean:1483`). For all states $\rho_1, \rho_2, \sigma_1, \sigma_2$ on $\mathbb{C}^d$ and all $p \in [0, 1]$, writing $\bar{\rho} := p\rho_1 + (1-p)\rho_2$ and $\bar{\sigma} := p\sigma_1 + (1-p)\sigma_2$,

$$D(\bar{\rho} \parallel \bar{\sigma}) \leq p D(\rho_1 \parallel \sigma_1) + (1-p) D(\rho_2 \parallel \sigma_2)$$

as an inequality in $[0, \infty]$.

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theorem qRelativeEnt_joint_convexity :
  ∀ (ρ₁ ρ₂ σ₁ σ₂ : MState d), ∀ (p : Prob),
    D(p ⌊ρ₁ ↔ ρ₂⌋ p ⌊σ₁ ↔ σ₂⌋) ≤ p * D(ρ₁ ⌋ σ₁) + (1 - p) * D(ρ₂ ⌋ σ₂)
```

4. Ingredients

The proof consumes two substantial pre-existing results (A, B), one elementary matrix lemma (C), and two scalar inequalities from Mathlib (D). Only Lemma E and the assembly in §5 are new in this contribution.

Proposition A (joint convexity of $\tilde{Q}_\alpha$, $\alpha > 1$; `sandwichedTraceFunctional_jointly_convex`, `DPI.lean:568`, pre-existing). Let $\alpha > 1$, let ι be a finite index set, let $w_i \geq 0$ with $\sum_i w_i = 1$, and let $(\rho_i), (\sigma_i)$ be states with $\text{supp } \rho_i \subseteq \text{supp } \sigma_i$ for every i . If $\bar{\rho}, \bar{\sigma}$ are states with $\bar{\rho} = \sum_i w_i \rho_i$ and $\bar{\sigma} = \sum_i w_i \sigma_i$ as matrices, then

$$\tilde{Q}_\alpha(\bar{\rho} \parallel \bar{\sigma}) \leq \sum_i w_i \tilde{Q}_\alpha(\rho_i \parallel \sigma_i).$$

Proof route (already formalized; sketch only). For $\alpha > 1$ and $\text{supp } \rho \subseteq \text{supp } \sigma$ one has the variational formula (`traceFunctional_eq_iSup_f_alpha`, `DPI.lean:434`)

$$\tilde{Q}_\alpha(\rho \parallel \sigma) = \sup_{H \geq 0} f_\alpha(H, \rho, \sigma), \quad f_\alpha(H, \rho, \sigma) = \alpha \text{Tr}[\rho H] - (\alpha - 1) \text{Tr}[(\sigma^{-\gamma} H \sigma^{-\gamma})^{\alpha/(\alpha-1)}]$$

(`f_alpha`, `DPI.lean:174`), the supremum being attained at $\hat{H} = \sigma^\gamma (\sigma^\gamma \rho \sigma^\gamma)^{\alpha-1} \sigma^\gamma$. For fixed $H \geq 0$, f_α is linear in ρ and convex in σ : with $s := -\gamma = \frac{\alpha-1}{2\alpha} \in (0, \frac{1}{2})$ and $q := \frac{\alpha}{\alpha-1} > 1$, the map $\sigma \mapsto \text{Tr}[(\sigma^s H \sigma^s)^q]$ is concave on the PSD cone (`HermitianMat.trace_conj_rpow_concave`, `QuantumInfo/ForMathlib/HermitianMat/LiebConcavity.lean:494`, itself derived from the Lieb–Ando trace theorems formalized in `QuantumInfo/ForMathlib/HayataGroup/TraceInequality/LiebAndoTrace.1` and it enters f_α with the negative coefficient $-(\alpha - 1)$. Hence each $f_\alpha(H, \cdot, \cdot)$ is jointly convex (`f_alpha_jointly_convex`, `DPI.lean:494`), and a pointwise supremum of jointly convex functions is jointly convex (`iSup_f_alpha_jointly_convex`, `DPI.lean:527`). This is the Frank–Lieb variational argument in the form used by Leditzky–Rouzé–Datta; see the literature note. $\square$

Proposition B (continuity in α ; `sandwichedRelEntropy_continuousOn`, `Relative.lean:1769`, pre-existing). For fixed states ρ, σ , the map $\alpha \mapsto \tilde{D}_\alpha(\rho \parallel \sigma)$ is continuous on $(0, \infty)$ (with respect to the order topology on $[0, \infty]$). The consequence used below is:

$$\tilde{D}_\alpha(\rho \parallel \sigma) \longrightarrow \tilde{D}_1(\rho \parallel \sigma) = D(\rho \parallel \sigma) \quad (\alpha \rightarrow 1^+).$$

Note that since $\tilde{D}_1 := D$ by definition, the $\alpha \rightarrow 1$ convergence of the sandwiched Rényi entropy to the Umegaki entropy is precisely the content of this continuity theorem; it is a genuine analytic fact proved earlier in the library, not a definitional triviality.

Lemma C (kernels of weighted sums; `HermitianMat.ker_weighted_sum_le`, `DPI.lean:545`). If $w_i \geq 0$, all $\rho_i, \sigma_i \geq 0$, and $\text{supp } \rho_i \subseteq \text{supp } \sigma_i$ for all i , then $\text{supp}(\sum_i w_i \rho_i) \subseteq \text{supp}(\sum_i w_i \sigma_i)$.

Scalar facts D (Mathlib). (i) $\log x \leq x - 1$ for $x > 0$ (`Real.log_le_sub_one_of_pos`). (ii) $|e^x - 1 - x| \leq x^2$ for $|x| \leq 1$ (`Real.abs_exp_sub_one_sub_id_le`); in particular $e^x - 1 \leq x + x^2$ there.

Also used: strict positivity $\tilde{Q}_\alpha(\rho \parallel \sigma) > 0$ under the support condition (`sandwichedTraceFunctional_pos`, `DPI.lean:100`).

5. The key new lemma: a convergent majorant for the difference quotient

Lemma E (`sandwichedTraceFunctional_sub_one_div_eventually_le`, `DPI.lean:1425`, `new`). Let ρ, σ be states with $\text{supp } \rho \subseteq \text{supp } \sigma$ (so $D(\rho\|\sigma) < \infty$). Then there is a function $u : \mathbb{R} \rightarrow \mathbb{R}$ with

$$u(\alpha) \xrightarrow{\alpha \rightarrow 1^+} D(\rho\|\sigma) \quad \text{and} \quad \frac{\tilde{Q}_\alpha(\rho\|\sigma) - 1}{\alpha - 1} \leq u(\alpha) \quad \text{for all } \alpha > 1 \text{ sufficiently close to } 1.$$

Proof. For $\alpha > 1$ put

$$r(\alpha) := \frac{\log \tilde{Q}_\alpha(\rho\|\sigma)}{\alpha - 1}, \quad x(\alpha) := (\alpha - 1)r(\alpha) = \log \tilde{Q}_\alpha(\rho\|\sigma),$$

and define $u(\alpha) := r(\alpha) + x(\alpha)r(\alpha) = r(\alpha) + (\alpha - 1)r(\alpha)^2$.

1. Under the support condition, $r(\alpha) \geq 0$ and $\tilde{D}_\alpha(\rho\|\sigma) = [r(\alpha)]^+$ (Definition 3 with `sandwichedRelEntropy_nonneg`), and $D(\rho\|\sigma) \neq \infty$. By Proposition B, $r(\alpha) \rightarrow D(\rho\|\sigma)$ (as real numbers) as $\alpha \rightarrow 1^+$ (Lean: `h_r_tendsto`, passing through `ENNReal.tendsto_toReal`).
2. Hence $x(\alpha) = (\alpha - 1)r(\alpha) \rightarrow 0$, so eventually $|x(\alpha)| \leq 1$ (Lean: `h_eps`, `h_small`).
3. $\tilde{Q}_\alpha(\rho\|\sigma) > 0$ (positivity above), so $\tilde{Q}_\alpha(\rho\|\sigma) = e^{x(\alpha)}$. By D(ii), for the eventual α of step 2, $\tilde{Q}_\alpha - 1 \leq x(\alpha) + x(\alpha)^2$; dividing by $\alpha - 1 > 0$,

$$\frac{\tilde{Q}_\alpha(\rho\|\sigma) - 1}{\alpha - 1} \leq r(\alpha) + x(\alpha)r(\alpha) = u(\alpha).$$

4. By steps 1–2, $u(\alpha) \rightarrow D(\rho\|\sigma) + 0 \cdot D(\rho\|\sigma) = D(\rho\|\sigma)$. ■

Heuristic remark (not part of the formal proof). Since $\tilde{Q}_1(\rho\|\sigma) = \text{Tr } \rho = 1$, the quantity $(\tilde{Q}_\alpha - 1)/(\alpha - 1)$ is a difference quotient whose two-sided limit “should” be $\frac{d}{d\alpha} \tilde{Q}_\alpha|_{\alpha=1} = D(\rho\|\sigma)$. The formal proof deliberately avoids establishing differentiability: a one-sided *eventual upper bound by a convergent majorant* suffices for the limit argument, and that follows from continuity of $\tilde{D}_\alpha$ alone, via the two elementary scalar bounds.

6. Proof of the theorem

Fix $\rho_1, \rho_2, \sigma_1, \sigma_2$ and p . The proof splits exactly as the Lean source does.

Step 0 — degenerate weights. If $p = 0$, then $\bar{\rho} = \rho_2$, $\bar{\sigma} = \sigma_2$ (in Lean via `Mixable.mix_zero`), and the right-hand side is $0 \cdot D(\rho_1\|\sigma_1) + 1 \cdot D(\rho_2\|\sigma_2) = D(\rho_2\|\sigma_2)$ — note this uses $0 \cdot \infty = 0$, so the case is genuine, not vacuous. Equality holds. If $p = 1$, symmetrically $\bar{\rho} = \rho_1$, $\bar{\sigma} = \sigma_1$ (the mixture identity $1 [x_1 \leftrightarrow x_2] = x_1$ is proved inline through injectivity of the underlying-matrix map, `Mixable.to_U_inj`), and the right side is $D(\rho_1\|\sigma_1)$. Henceforth $0 < p < 1$ (Lean: `hp0'`, `hp1'`).

Step 1 — infinite right-hand side. Suppose $\text{supp } \rho_1 \not\subseteq \text{supp } \sigma_1$ (Lean: `hker_1` fails). Then $D(\rho_1\|\sigma_1) = \infty$ by Definition 1, and since $p \neq 0$, $p \cdot \infty = \infty$, so the right-hand side is ∞ and the inequality is trivial (`1e_top`). The case $\text{supp } \rho_2 \not\subseteq \text{supp } \sigma_2$ is identical, using $1 - p \neq 0$. So we may assume both support conditions

$$\text{supp } \rho_i \subseteq \text{supp } \sigma_i \quad (i = 1, 2),$$

whence $D(\rho_1\|\sigma_1), D(\rho_2\|\sigma_2) < \infty$ (Lean: `h_ne1`, `h_ne2`).

Step 2 — setup for the main case. Write the mixtures as weighted sums over $\iota = \{1, 2\}$ (Lean: `Fin 2`) with weight vector $w = (p, 1 - p)$:

$$\bar{\rho} = \sum_i w_i \rho_i, \quad \bar{\sigma} = \sum_i w_i \sigma_i$$

as matrix identities (Lean: `hM_rho`, `hM_sigma`, by unfolding `Mixable.mix`); $w_i \geq 0$ and $\sum_i w_i = 1$ (`hw_nonneg`, `hw_sum`). By Lemma C, $\text{supp } \bar{\rho} \subseteq \text{supp } \bar{\sigma}$ (Lean: `hker_mix`). Apply Lemma E to each pair to obtain majorants u_1, u_2 with $u_i(\alpha) \rightarrow D(\rho_i\|\sigma_i)$ as $\alpha \rightarrow 1^+$ and, eventually, $(\tilde{Q}_\alpha(\rho_i\|\sigma_i) - 1)/(\alpha - 1) \leq u_i(\alpha)$.

Step 3 — pointwise bound for $\alpha > 1$ (Lean: `h_ev`). For $\alpha > 1$ in the eventual range of both majorants, using $\tilde{Q}_\alpha(\bar{\rho}\|\bar{\sigma}) > 0$ (positivity, with `hker_mix`) and Definition 3:

$$\tilde{D}_\alpha(\bar{\rho}\|\bar{\sigma}) = \left\lceil \frac{\log \tilde{Q}_\alpha(\bar{\rho}\|\bar{\sigma})}{\alpha - 1} \right\rceil^+,$$

and we bound the bracketed real number:

$$\begin{aligned} \frac{\log \tilde{Q}_\alpha(\bar{\rho}\|\bar{\sigma})}{\alpha - 1} &\leq \frac{\tilde{Q}_\alpha(\bar{\rho}\|\bar{\sigma}) - 1}{\alpha - 1} && [\text{D(i): } \log x \leq x - 1] \\ &\leq \frac{p \tilde{Q}_\alpha(\rho_1\|\sigma_1) + (1-p) \tilde{Q}_\alpha(\rho_2\|\sigma_2) - 1}{\alpha - 1} && [\text{Proposition A, } \iota = \{1, 2\}] \\ &= p \frac{\tilde{Q}_\alpha(\rho_1\|\sigma_1) - 1}{\alpha - 1} + (1-p) \frac{\tilde{Q}_\alpha(\rho_2\|\sigma_2) - 1}{\alpha - 1} && [\text{since } p + (1-p) = 1] \\ &\leq p u_1(\alpha) + (1-p) u_2(\alpha) && [\text{Lemma E}]. \end{aligned}$$

Hence, eventually as $\alpha \rightarrow 1^+$,

$$\tilde{D}_\alpha(\bar{\rho}\|\bar{\sigma}) \leq \left\lceil p u_1(\alpha) + (1-p) u_2(\alpha) \right\rceil^+.$$

Step 4 — pass to the limit $\alpha \rightarrow 1^+$. By Proposition B (applied to the mixture pair), the left side tends to $D(\bar{\rho}\|\bar{\sigma})$ (Lean: `h_lhs`). By Lemma E and continuity of $t \mapsto [t]^+$, the right side tends to

$$\left\lceil p D(\rho_1\|\sigma_1) + (1-p) D(\rho_2\|\sigma_2) \right\rceil^+ = p D(\rho_1\|\sigma_1) + (1-p) D(\rho_2\|\sigma_2)$$

(Lean: `h_rhs`; the identification of the $[\cdot]^+$ of the real convex combination with the $[0, \infty]$ -valued convex combination uses the finiteness from Step 1, via `ENNReal.ofReal_toReal` and friends). Since the inequality of Step 3 holds eventually along the (nontrivial) filter of right-neighbourhoods of 1, the inequality passes to the limits (`le_of_tendsto_of_tendsto`):

$$D(\bar{\rho}\|\bar{\sigma}) \leq p D(\rho_1\|\sigma_1) + (1-p) D(\rho_2\|\sigma_2). \quad \blacksquare$$

7. Lean-paper dictionary

Paper object / step	Lean identifier	Location (commit 9385c7f3)
Umegaki D	<code>qRelativeEnt</code> / D	<code>Relative.lean:1497</code>

Paper object / step	Lean identifier	Location (commit 9385c7f3)
Sandwiched $\tilde{D}_\alpha$	<code>SandwichedRelEntropy / D_α</code>	<code>Relative.lean:1480</code>
Trace functional $\tilde{Q}_\alpha$	<code>sandwichedTraceFunctional / Q_α</code>	<code>DPI.lean:66</code>
$\tilde{D}_\alpha = [\log \tilde{Q}_\alpha / (\alpha - 1)]^+$	<code>sandwichedRelEntropy_eq_logDPI.lean:78</code>	<code>Relative.lean:1480</code>
Proposition A	<code>sandwichedTraceFunctional_jointly_e</code>	<code>DPI.lean:568</code>
— variational formula	<code>traceFunctional_eq_iSup_f_alpha, H_hat</code>	<code>Relative.lean:434, 174, 180</code>
— concavity in σ	<code>HermitianMat.trace_conj_rpow_LiebConcavity</code>	<code>Relative.lean:494</code>
Proposition B	<code>sandwichedRelEntropy.continuous</code>	<code>Relative.lean:1769</code>
Lemma C	<code>HermitianMat.ker_weighted_sum</code>	<code>DPI.lean:545</code>
D(i)	<code>Real.log_le_sub_one_of_pos</code>	<code>Mathlib</code>
D(ii)	<code>Real.abs_exp_sub_one_sub_id</code>	<code>Mathlib</code>
$\tilde{Q}_\alpha > 0$	<code>sandwichedTraceFunctional_pos</code>	<code>DPI.lean:100</code>
Lemma E	<code>sandwichedTraceFunctional_sub_one_div_1</code>	<code>DPI.lean:1425</code>
Steps 0–4	<code>qRelativeEnt_joint_convexity</code>	<code>DPI.lean:1483</code>
Step 3	<code>h_ev (with h_convex)</code>	<code>DPI.lean:1572–1594</code>
Step 4	<code>h_lhs, h_rhs, le_of_tendsto_of_tendsto</code>	<code>DPI.lean:1548–1595</code>

8. Remarks for the reviewer

- **What is new.** Lemma E and the theorem itself (§5–§6), ~185 lines added to `DPI.lean`; the previously `@[sorryful]` statement was removed from `Relative.lean`. Propositions A and B and all deeper inputs (Lieb–Ando trace theorems) were already in the library. The theorem was placed in `DPI.lean` rather than `Relative.lean` because the $\tilde{Q}_\alpha$ machinery lives there and importing it from `Relative.lean` would create a cycle.
- **No differentiability is needed.** The classical way to see $\tilde{D}_\alpha \rightarrow D$ and to transfer convexity is to differentiate $\alpha \mapsto \tilde{Q}_\alpha$ at $\alpha = 1$; the formal proof replaces this by the one-sided majorant of Lemma E, which needs only continuity (Proposition B) plus the scalar bounds D. This is the main proof-engineering choice.
- **The statement is the binary form.** Finite mixtures of n states follow by induction in the usual way, but only the n -ary form of Proposition A (arbitrary finite `Fintype ι`) is formalized; the `D`-level statement is for two pairs, matching the pre-existing `sorryful` statement it closes.
- **Axioms.** The proof is `sorry`-free; `#print axioms qRelativeEnt_joint_convexity` reports only `propext`, `Classical.choice`, `Quot.sound` (recorded in the commit message of 9385c7f3).